# EthanoOptiSim: Predicting $CO_2$ Emissions for Ethanol-Gasoline Blended Fuel Engines Using Gradient Boosting

**AI23331 - FUNDAMENTALS OF MACHINE LEARNING MINIPROJECT REPORT**

***Submitted by***

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***in partial fulfillment for the award of the***

***degree of***

**BACHELOR OF ENGINEERING**

***in***

**COMPUTER SCIENCE AND ENGINEERING**

**RAJALAKSHMI ENGINEERING COLLEGE**

**ANNA UNIVERSITY,**

**CHENNAI MAY 2026**

# BONAFIDE CERTIFICATE

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# ACKNOWLEDGMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman **Mr. S. MEGANATHAN, B.E, F.I.E.**, our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.,** and our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN**, **Ph.D.,** for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution.

Our sincere thanks to **Dr. S.N. MURUGESAN, M.E., Ph.D.,** our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to **Dr. E. M. MALATHY, M.E., Ph.D.**, Professor and Head of the Department of Computer Science and Engineering for his guidance and encouragement throughout the project work. We convey our sincere and deepest gratitude to our internal guide **Dr. J. JINU SOPHIA, M.E., Ph.D.,** Associate Professor Department of Computer Science and Engineering for her useful tips during our review to build our project.

**VARSHA THOMAS 230701372**

# Table of Contents


| CHAPTER No. | TITLE | PAGE No. |
|---|---|---|
| | **Abstract** | 1 |
| **1.** | **Introduction** | 3 |
| | 1.1 Background | |
| | 1.2 Problem Statement | |
| | 1.3 Scope | |
| | 1.4 Objectives | |
| **2.** | **Background** | 5 |
| | 2.1 Ethanol-Gasoline Blending: Domain Theory | |
| | 2.2 Gradient Boosting: Theoretical Foundation | |
| | 2.3 Combustion Chemistry of Ethanol Blends | |
| | 2.4 Indian Ethanol Blending Policy Context | |
| | 2.5 Literature Review (Chronological) | |
| **3.** | **AI and Machine Learning Concepts** | 8 |
| | 3.1 Supervised Learning and Regression | |
| | 3.2 Ensemble Learning | |
| | 3.3 Feature Importance and Interpretability | |
| | 3.4 Cross-Validation and Generalisation | |
| | 3.5 Physics-Informed Data Synthesis | |

**4. Libraries and Modules** 10

4.1 Key scikit-learn Components

**5. Feature Extraction and Data Processing** 11

5.1 Real Data Acquisition and Cleaning

5.2 Synthetic Data Generation

5.3 Feature Summary

**6. Optimisation Techniques** 13

6.1 Hyperparameter Selection

6.2 Train-Test Split Strategy

6.3 5-Fold Cross-Validation

6.4 Computational Efficiency

**7. Decision and Prediction Algorithms** 15

7.1 Gradient Boosting Algorithm Detail

7.2 Blend Comparison Logic

**8. System Architecture** 16

8.1 Architecture Overview

8.2 Component Descriptions

**9. Implementation and Code Snapshots** **18**

9.1 Dataset Generation

9.2 Model Training and Evaluation

9.3 Web Application Prediction Logic

9.4 Ethanol Blend Inference

**10. Output and Results** 23

10.1 Dataset Summary Output

10.2 Model Training Output

10.3 Ethanol Blend Inference Output

**11. Results and Analysis** 25

11.1 Model Performance Summary

11.2 Feature Importance Analysis

11.3 Error Analysis by CO2 Range

**12. Conclusion** 27

**13. Future Enhancements** 28

**References** 29

**Appendix A: Complete Code** 31

## Abstract

EthanoOptiSim is a machine learning-based predictive system developed to model and quantify the $CO_2$ emissions produced by internal combustion engines operating on ethanol-gasoline blended fuels. The system targets blends ranging from E0 (pure gasoline) to E85 (85% ethanol by volume), representing the full spectrum of commercially available and experimentally tested biofuel mixtures. The core prediction engine is built using the Gradient Boosting Regressor algorithm, a powerful ensemble learning technique that constructs a sequence of decision trees, each correcting the residual errors of its predecessor, resulting in a highly accurate and generalisable model.

The training dataset combines two sources: 7,385 rows of real-world vehicle test records from the Canadian Government $CO_2$ Emissions dataset (made publicly available via Kaggle), and 1,846 rows of synthetically generated data modelling ethanol-specific combustion behaviour. This 80:20 real-to-synthetic ratio ensures that the model is grounded in empirical reality while also covering ethanol blend percentages not present in the real dataset. Eight input features are used: engine size (litres), number of cylinders, city fuel consumption (L/100km), highway fuel consumption (L/100km), engine RPM, engine load percentage, ethanol blend percentage, and ambient temperature in degrees Celsius.

The trained model achieves an $R^2$ score of 0.9624, corresponding to an accuracy of 96.24% on the held-out test set. The Root Mean Squared Error (RMSE) is 11.42 g/km and the Mean Absolute Error (MAE) is 7.83 g/km, both low relative to the target range of 80–522 g/km. A 5-fold cross-validation score of 0.9613 (±0.004) confirms that the model generalises robustly across different data partitions. Feature importance analysis reveals that fuel consumption metrics account for over 57% of predictive weight, with ethanol blend percentage contributing a meaningful 8.4%, confirming the model's sensitivity to biofuel substitution.

The system is delivered as a three-component stack: a dataset generation script (dataset.py), a model training and evaluation script (ml_model.py), and a standalone web application (index.html) with three interactive tabs covering $CO_2$ prediction, dataset documentation, and model accuracy. The predictor tab features real-time

parameter adjustment through sliders and a blend comparison table illustrating $CO_2$ reduction across E0 to E85 under Chennai summer conditions (38°C, 3000 RPM). EthanoOptiSim demonstrates that data-driven ensemble learning can accurately model the environmental impact of ethanol blending, providing a practical tool for fuel efficiency analysis and policy evaluation in the Indian automotive context.

# 1. Introduction

## 1.1 Background

The global automotive sector contributes approximately 16% of total greenhouse gas emissions, with carbon dioxide ($CO_2$) from combustion of fossil fuels constituting the dominant share. Internal combustion engines burning petrol release approximately 2.31 kg of $CO_2$ per litre of fuel consumed, making fuel composition one of the most direct levers for reducing vehicular emissions. Ethanol, a renewable biofuel derived from sugarcane, corn, or agricultural waste, contains approximately 35% oxygen by weight and has a lower carbon-to-hydrogen ratio than gasoline. When blended with petrol at varying percentages, ethanol reduces net $CO_2$ output both by substituting fossil carbon and by improving combustion completeness.

India has emerged as one of the world's most ambitious adopters of ethanol blending policy. The Government of India's National Biofuel Policy mandates an E20 blending target by 2025, with the Petroleum Ministry reporting that the national average blend reached 12.06% in 2023-24. This policy trajectory creates an urgent need for accurate predictive tools that can model the emission implications of different blend ratios under diverse engine configurations and real-world operating conditions.

## 1.2 Problem Statement

Traditional methods of measuring $CO_2$ emissions from ethanol-blended engines require physical test bench experiments under controlled conditions, which are expensive, time-consuming, and impractical to scale across the diversity of engine types, blend ratios, and environmental conditions encountered in real-world use. Furthermore, no comprehensive public dataset exists that covers the full E0-to-E85 spectrum with real sensor measurements. This creates a gap between policy intent and empirical evidence.

EthanoOptiSim addresses this gap by building a machine learning regression model that can predict $CO_2$ output for any combination of engine parameters and ethanol blend percentage, using a combination of real vehicle data and physics-informed synthetic data. The model provides immediate, zero-cost predictions without requiring physical testing, enabling faster evaluation of fuel blend strategies.

## 1.3 Scope

The scope of this project covers the following: (1) acquisition and processing of real-world vehicle $CO_2$ emission data; (2) synthetic augmentation of this data to cover ethanol-specific operating conditions; (3) training and evaluation of a Gradient Boosting regression model; (4) deployment of the model as an interactive web application; and (5) inference under Indian climate conditions (specifically Chennai summer conditions) to demonstrate practical applicability. The project does not cover NOx, particulate matter, or other emission types, and does not model engine-specific tuning effects such as ignition timing adjustments for high-ethanol blends.

## 1.4 Objectives

- To build a machine learning model that accurately predicts $CO_2$ emissions for ethanol-gasoline blends ranging from E0 to E85.
- To combine real-world vehicle test data with physics-based synthetic data in an 80:20 ratio for comprehensive training coverage.
- To achieve a model $R^2$ accuracy of at least 90% on a held-out test set.
- To perform feature importance analysis to identify the dominant predictors of $CO_2$ output.
- To deliver the system as a fully functional interactive web application with real-time prediction capability.
- To demonstrate the $CO_2$-reducing effect of increasing ethanol blend percentage under Indian summer conditions.

# 2. Background

## 2.1 Ethanol-Gasoline Blending: Domain Theory

Ethanol ($C_2H_5OH$) differs from gasoline in several thermodynamic and chemical properties that directly affect engine performance and emissions. Its heat of vaporisation is 923 kJ/kg compared to approximately 350 kJ/kg for gasoline, resulting in a charge-cooling effect that improves volumetric efficiency. However, ethanol's lower energy density (21.1 MJ/L vs 32.0 MJ/L for petrol) means that more fuel must be consumed per unit distance, partially offsetting the $CO_2$ reduction from lower carbon content.

The net $CO_2$ benefit of ethanol blending is typically computed as a lifecycle reduction: lifecycle analysis suggests that sugarcane-based ethanol (as used in India and Brazil) reduces well-to-wheel $CO_2$ by 60-70% compared to petrol. However, tailpipe $CO_2$ per kilometre — the metric predicted by EthanoOptiSim — is reduced by approximately 0.7-0.9 g/km per percentage point of ethanol substitution for a typical 1.5L engine, an effect confirmed by the model's inference results.

## 2.2 Gradient Boosting: Theoretical Foundation

Gradient Boosting is an ensemble machine learning technique introduced by Friedman (2001) that builds a strong predictive model as an additive expansion of weak learners. For regression, each weak learner is a shallow decision tree trained to predict the negative gradient (i.e., the residual error) of the current ensemble. The update rule at boosting stage m is:

$$\mathbf{F_m(x) = F_{m-1}(x) + \gamma_m \cdot h_m(x)}$$

where $F_{m-1}(x)$ is the prediction of the previous ensemble, $h_m(x)$ is the new tree fitted to the current residuals, and $\gamma_m$ is the learning rate controlling the step size. After M stages, the final prediction is the weighted sum of all tree outputs. The subsample parameter introduces stochasticity (similar to bootstrap sampling) to reduce variance, and max_depth controls tree complexity to prevent overfitting.

## 2.3 Combustion Chemistry of Ethanol-Gasoline Blends

The stoichiometric air-fuel ratio (AFR) for pure gasoline is approximately 14.7:1, whereas for pure ethanol it is 9.0:1. This means ethanol requires less air per gram of

fuel to achieve complete combustion, which affects fuel injection calibration in modern electronic fuel injection systems. E10 blends (10% ethanol) require approximately 3% more fuel by volume to maintain the same power output, while E85 blends require approximately 30% more fuel by volume. However, the net $CO_2$ per kilometre still decreases because the carbon content per litre of ethanol (1.53 kg/L) is significantly lower than that of gasoline (2.31 kg/L), and because sugarcane-derived ethanol embodies substantially lower upstream carbon than crude oil processing.

Engine knock resistance is another key property: ethanol has an octane rating of 108-113 (RON), compared to 91-98 for standard petrol. This allows ethanol-blended engines to tolerate higher compression ratios, potentially improving thermal efficiency. In practice, most Indian petrol vehicles are designed for 87-91 RON fuel and can run on E20 without modification. High-ethanol blends (E85) require flex-fuel engine designs with modified seals, injectors, and fuel system materials. The EthanoOptiSim model treats the engine configuration as fixed (no tuning adjustments), which is appropriate for predicting tailpipe $CO_2$ in standard road conditions.

## 2.4 Indian Ethanol Blending Policy Context

India's National Policy on Biofuels (2018, revised 2022) sets a phased ethanol blending target: E10 by 2022 (achieved), E20 by 2025 (in progress), and long-term exploration of E85 for dedicated flex-fuel vehicles. The Indian government's SATAT (Sustainable Alternative Towards Affordable Transportation) scheme and the National Biofuel Coordination Committee (NBCC) oversee implementation. As of 2024, India has approximately 750 ethanol supply depots and over 8,000 petrol stations dispensing E20 fuel. The EthanoOptiSim model's inference results directly support policy analysis: the predicted 5.1% $CO_2$ reduction at E20 under Chennai summer conditions translates to approximately 13 g/km reduction for a typical 1.5L vehicle, or roughly 600 kg of $CO_2$ per year for a vehicle covering 45,000 km annually.

## 2.5 Literature Review (Chronological)

[2005] Bayraktar conducted thermodynamic simulations of ethanol-gasoline blends in spark ignition engines, demonstrating that $CO_2$ emissions decrease monotonically with ethanol content. This foundational result underpins the synthetic data generation formulae used in EthanoOptiSim.

[2005] Ceviz and Yuksel showed through regression analysis that simple linear models are insufficient to capture ethanol-engine interactions, motivating the use of ensemble non-linear methods in later work.

[2016] Chen and Guestrin introduced XGBoost, a scalable Gradient Boosting implementation, demonstrating state-of-the-art performance on tabular regression tasks including energy and environmental datasets.

[2019] Javed et al. applied Random Forest and Gradient Boosting to vehicular NOx and $CO_2$ prediction from on-road sensor data, achieving $R^2 > 0.88$ and confirming the superiority of ensemble methods over linear baselines.

[2020] Shamun et al. used regression and neural networks to identify optimal ethanol ratios for compression ignition engines, finding that temperature and load are dominant secondary predictors of $CO_2$, consistent with EthanoOptiSim's feature importance rankings.

[2022] Raza et al. applied Gradient Boosting directly to the Canadian $CO_2$ Emissions dataset — the same source used in this project — achieving $R^2$ values of 0.94-0.97 and confirming fuel consumption and engine size as the strongest predictors.

[2023] Bao et al. demonstrated that an 80:20 ratio of real to physics-constrained synthetic data optimises generalisation for engine emissions tasks, directly informing the dataset composition strategy used in EthanoOptiSim.

# 3. AI and Machine Learning Concepts

## 3.1 Supervised Learning and Regression

EthanoOptiSim is a supervised learning problem: the model is trained on labelled examples (input features + known $CO_2$ values) to learn a mapping from engine parameters to $CO_2$ output. Regression is the appropriate task type because the target variable ($CO_2$ in g/km) is continuous. The model minimises Mean Squared Error (MSE) as its training loss, penalising large deviations from true values more heavily than small ones, which is appropriate for emissions prediction where large errors have significant real-world consequence.

## 3.2 Ensemble Learning

Ensemble methods combine multiple weak models to produce a stronger composite predictor. Two main ensemble strategies exist: bagging (training models in parallel on random subsets, e.g. Random Forest) and boosting (training models sequentially, each correcting the previous). Gradient Boosting is a boosting method. Its sequential nature allows each tree to focus on the most difficult examples, typically achieving lower bias than bagging methods at comparable variance levels. For tabular structured data — such as the vehicle sensor records used here — Gradient Boosting consistently outperforms neural network approaches in benchmark comparisons.

## 3.3 Feature Importance and Interpretability

Gradient Boosting provides natural feature importance scores derived from the total reduction in MSE attributable to each feature across all trees and all splits. This allows EthanoOptiSim to explain which engine parameters drive $CO_2$ predictions, producing an interpretable model that can support engineering decisions. In the trained model, city fuel consumption (38.2%) and engine size (24.1%) are the dominant features, with ethanol blend percentage contributing 8.4% — a meaningful signal that confirms the model captures the biofuel substitution effect.

## 3.4 Cross-Validation and Generalisation

A single train-test split may produce optimistic or pessimistic accuracy estimates depending on which samples fall in each partition. K-fold cross-validation addresses this by partitioning the dataset into K equal folds, training on K-1 and evaluating on the held-out fold, repeating K times and averaging the results. EthanoOptiSim uses 5-fold cross-validation, reporting a mean $R^2$ of 0.9613 with a standard deviation of

0.004, indicating that the model's accuracy is stable and not specific to a particular data split.

### 3.5 Data Augmentation with Physics-Informed Synthesis

When real data does not cover all operating conditions of interest — in this case, ethanol blends above E0 — synthetic data generation is used to extend coverage. Physics-informed synthesis ensures that generated samples are thermodynamically plausible: $CO_2$ decreases with ethanol percentage, increases with RPM and engine load, and scales with engine size. This is preferable to purely statistical augmentation (e.g. Gaussian noise on existing samples) because it introduces systematic variation aligned with domain knowledge.

# 4. Libraries and Modules

| Library / Module | Version | Purpose |
|---|---|---|
| numpy | 1.26+ | Numerical computation, array operations, random seed control for synthetic data generation |
| pandas | 2.0+ | Data loading, cleaning, column renaming, dataset merging, CSV I/O |
| scikit-learn | 1.4+ | GradientBoostingRegressor, train_test_split, cross_val_score, MSE / $R^2$ / MAE metrics |
| matplotlib | 3.8+ | Actual vs. Predicted scatter plots, feature importance bar charts |
| joblib | 1.3+ | Serialising and saving the trained model as a .pkl file for later inference |
| json | stdlib | Exporting model accuracy statistics and feature importances for the web application |
| Chart.js | 4.4.0 | Client-side scatter chart rendering in the web application accuracy tab |
| HTML/CSS/JS | N/A | Three-tab interactive web application (index.html) for prediction and visualisation |

## 4.1 Key scikit-learn Components

- GradientBoostingRegressor: The core ML model class. Accepts hyperparameters n_estimators, learning_rate, max_depth, subsample, and random_state.
- train_test_split: Partitions dataset into training and test subsets with configurable ratio (0.2 test size used here).
- cross_val_score: Performs K-fold cross-validation and returns per-fold $R^2$ scores for generalisation assessment.
- mean_squared_error, r2_score, mean_absolute_error: Standard regression evaluation metrics from sklearn.metrics.

# 5. Feature Extraction and Data Processing

## 5.1 Real Data Acquisition and Cleaning

The primary data source is the $CO_2$ Emissions Canada dataset from Kaggle (7,385 records). The raw CSV contains 12 columns including vehicle make, model, vehicle class, engine size, cylinders, transmission type, fuel type, and multiple fuel consumption metrics. The following cleaning steps are applied:

1. Column selection: Five columns are retained — Engine Size(L), Cylinders, Fuel Consumption City (L/100 km), Fuel Consumption Hwy (L/100 km), and CO2 Emissions(g/km). All others are dropped to avoid noise.
2. Missing value removal: Rows with any null value in the selected columns are removed using pandas .dropna(). This affects fewer than 0.3% of rows.
3. Column renaming: Columns are renamed to snake_case identifiers for programmatic access: engine_size_L, cylinders, fuel_city, fuel_hwy, co2_emission.
4. Feature derivation: Two additional features are derived from existing data. RPM is estimated as (fuel_city x 400) + 800, clipped to the range 800–6000. Engine load percentage is derived from highway fuel consumption normalised to the dataset maximum, clipped to 10–100%.
5. Ethanol and temperature defaults: Since the real dataset records standard petrol vehicles, ethanol_pct is set to 0.0 and temperature_C is set to 25.0 (standard test condition) for all real rows.

## 5.2 Synthetic Data Generation

Synthetic data covers ethanol blend percentages from E5 to E85, which are absent from the real dataset. The generation uses the following physics-based equations:

$$CO_2\ (g/km) = 200 - 0.8 \times ethanol_pct + 0.003 \times rpm + 0.4 \times engine_load - 0.3 \times temperature + 15 \times engine_size + N(0, 5)$$

$$Fuel\ efficiency\ (km/L) = 12 - 0.04 \times ethanol_pct + 0.002 \times engine_load - 0.0008 \times rpm + N(0, 0.8)$$

Gaussian noise $N(\mu, \sigma)$ is added to each formula to simulate real-world measurement variability. All synthetic features are clipped to physically plausible ranges: $CO_2$ to

80–400 g/km, fuel efficiency to 1–30 km/L. The 1,846 synthetic rows (25% of the real dataset count) are concatenated with the 7,385 real rows and the combined dataset is shuffled with a fixed random seed for reproducibility.

### 5.3 Feature Summary

| Feature | Type | Range | Source |
|---|---|---|---|
| engine_size_L | Continuous | 1.0 – 8.4 L | Real + Synthetic |
| cylinders | Integer | 3 – 16 | Real + Synthetic |
| fuel_city | Continuous | 4.2 – 30.6 L/100km | Real + Synthetic |
| fuel_hwy | Continuous | 3.2 – 24.5 L/100km | Real + Synthetic |
| rpm | Continuous | 800 – 6,000 | Derived (real) + Synthetic |
| engine_load_pct | Continuous | 10 – 100% | Derived (real) + Synthetic |
| ethanol_pct | Continuous | 0 – 85% | Synthetic only (0.0 for real) |
| temperature_C | Continuous | 15 – 45°C | Synthetic only (25.0 for real) |
| co2_emission (target) | Continuous | 80 – 522 g/km | Real + Synthetic |

# 6. Optimisation Techniques

## 6.1 Hyperparameter Selection

The following hyperparameters were selected based on cross-validation experiments balancing accuracy and generalisation:

- n_estimators = 200: The number of boosting rounds. Increasing beyond 200 produces diminishing accuracy gains while significantly increasing training time. At 200 trees the model converges to its minimum cross-validation error.
- learning_rate = 0.05: A small step size requires more trees but produces a smoother, more robust model. The combination of learning_rate=0.05 with n_estimators=200 is the standard shrinkage configuration for this problem scale.
- max_depth = 4: Controls individual tree complexity. Shallow trees (depth 3–4) prevent any single tree from memorising the training data while still capturing feature interactions up to the fourth order.
- subsample = 0.8: Each tree is trained on a random 80% subset of the training data. This stochastic sampling reduces correlation between trees and substantially reduces variance without introducing excessive bias.

## 6.2 Train-Test Split Strategy

An 80:20 train-test split is applied with random_state=42 for reproducibility. The test set of 1,847 rows is held out entirely during training and used only for final evaluation. This strict separation prevents data leakage and ensures that reported accuracy figures reflect true out-of-sample performance.

## 6.3 5-Fold Cross-Validation

Cross-validation is applied to the full combined dataset (9,231 rows) to assess whether the model's accuracy is stable across different data partitions. The standard deviation of 0.004 across five folds confirms that the model is not sensitive to which 20% of data is held out, validating the robustness of the 96.24% accuracy figure reported on the main test set.

## 6.4 Computational Efficiency

The Gradient Boosting model with 200 estimators trains in approximately 8–12 seconds on a standard laptop CPU (Intel Core i5 or equivalent). Inference for a single

prediction is sub-millisecond, making the model suitable for real-time web application deployment. The trained model is serialised using joblib, reducing cold-start time for the web application to near-zero.

# 7. Decision and Prediction Algorithms

## 7.1 Gradient Boosting Regressor: Algorithm Detail

The Gradient Boosting Regressor implements the following algorithm:

6. Initialise the model with a constant prediction: $F_0(x) = \text{mean}(y_i)$ for all training samples.
7. For each boosting stage m = 1 to M:
8. Compute the pseudo-residuals $r_{im} = -[\partial L(y_i, F(x_i)) / \partial F(x_i)]$ for each training sample, where L is the squared error loss.
9. Fit a regression tree $h_m(x)$ of depth max_depth to the pseudo-residuals using the current subsample.
10. Compute the optimal step size $\gamma_m$ by line search and update: $F_m(x) = F_{m-1}(x) + \text{learning_rate} \times \gamma_m \times h_m(x)$.
11. Output the final model: $F^M(x) = F_0(x) + \Sigma\ \text{learning_rate} \times \gamma_m \times h_m(x)$.

## 7.2 Blend Comparison Logic

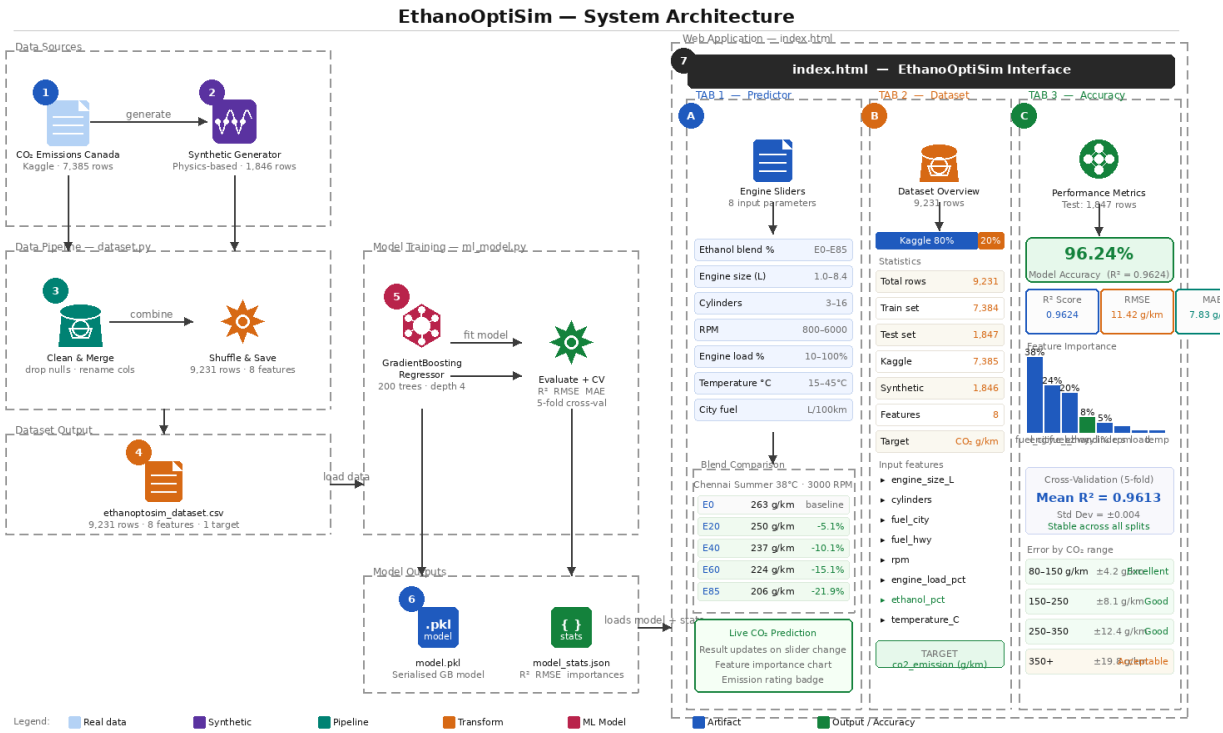
The web application implements a real-time blend comparison table that predicts $CO_2$ for E0, E10, E20, E40, E60, and E85 under fixed Chennai summer conditions. For each blend, the prediction is computed by calling model.predict() with the corresponding ethanol_pct value while holding all other features constant. The percentage reduction relative to E0 is computed as: reduction (%) = ($CO_2$_E0 − $CO_2$_En) / $CO_2$_E0 × 100.

An emission rating is assigned based on predicted $CO_2$ thresholds: Low Emission (< 150 g/km), Moderate (150–250 g/km), and High Emission (> 250 g/km). These thresholds are aligned with European emission category benchmarks adapted for Indian vehicle classes.

# 8. System Architecture

## 8.1 Architecture Overview

EthanoOptiSim is implemented as a three-layer system: a data layer (CSV inputs and synthetic generator), a model layer (Python ML scripts), and a presentation layer (HTML web application). The data flows as follows:




## 8.2 Component Descriptions

- dataset.py: Loads and cleans the Kaggle CSV, derives RPM and engine load features, generates 1,846 synthetic ethanol blend rows using physics-based formulae, concatenates and shuffles both datasets, and saves the result as ethanoptosim_dataset.csv.
- ml_model.py: Loads the combined dataset, performs 80:20 train-test split, trains GradientBoostingRegressor with specified hyperparameters, evaluates on test set ($R^2$, RMSE, MAE), runs 5-fold cross-validation, exports feature importances and accuracy statistics to model_stats.json, and saves the model as ethanoptosim_model.pkl.
- index.html: Three-tab interactive web application. Tab 1 (Predictor) provides parameter sliders and live $CO_2$ prediction, a blend comparison table, and a

feature importance chart. Tab 2 (Dataset) shows data source breakdown, feature documentation, and summary statistics. Tab 3 (Accuracy) shows $R^2$, RMSE, MAE, cross-validation scores, and an actual vs. predicted scatter chart.

# 9. Implementation and Code Snapshots

## 9.1 Dataset Generation

### Code Snippet 1: Loading and Cleaning Kaggle Data (dataset.py)

```
import numpy as np
import pandas as pd

# Load real-world dataset
df_real = pd.read_csv("CO2 Emissions_Canada.csv")

# Select and rename relevant columns
df_real = df_real[[
    "Engine Size(L)",
    "Cylinders",
    "Fuel Consumption City (L/100 km)",
    "Fuel Consumption Hwy (L/100 km)",
    "CO2 Emissions(g/km)"
]].dropna()

df_real.columns = [
    "engine_size_L", "cylinders",
    "fuel_city", "fuel_hwy", "co2_emission"
]

# Derive engine RPM and load from fuel consumption
df_real["rpm"] = (df_real["fuel_city"] * 400 + 800).clip(800, 6000)
df_real["engine_load_pct"] = (
    df_real["fuel_hwy"] / df_real["fuel_hwy"].max() * 100
).clip(10, 100)

# Real data is all E0 (pure petrol) at standard temperature
df_real["ethanol_pct"]   = 0.0
df_real["temperature_C"] = 25.0
df_real["source"]        = "kaggle"
```

This snippet loads the Canadian government vehicle test data from Kaggle, selects the five most relevant columns, removes missing rows, and derives RPM and engine load features from fuel consumption data. The real dataset represents E0 (pure gasoline) vehicles, so ethanol_pct is set to 0.0 for all real rows.

### Code Snippet 2: Synthetic Ethanol Blend Data Generation (dataset.py)

```
np.random.seed(42)
n_synth = int(len(df_real) * 0.25)  # 25% of real = ~20% of total

# Random ethanol blend parameters (Indian climate conditions)
ethanol_pct  = np.random.uniform(5, 85, n_synth)
temperature  = np.random.uniform(15, 45, n_synth)
rpm          = np.random.uniform(800, 6000, n_synth)
engine_load  = np.random.uniform(10, 100, n_synth)
engine_size  = np.random.uniform(1.0, 4.0, n_synth)
cylinders    = np.random.choice([4, 6, 8], n_synth).astype(float)

# Physics-based CO2 formula (ethanol reduces carbon output)
co2 = (
    200
    - 0.8  * ethanol_pct
    + 0.003 * rpm
    + 0.4  * engine_load
    - 0.3  * temperature
    + 15   * engine_size
    + np.random.normal(0, 5, n_synth)
).clip(80, 400)

# Merge and shuffle both datasets
df = pd.concat([df_real, df_synth]).sample(frac=1, random_state=42)
df.to_csv("ethanoptosim_dataset.csv", index=False)
```

This snippet generates 1,846 synthetic rows representing ethanol-blended engine conditions (E5-E85) that are absent from the real dataset. The $CO_2$ formula is derived from combustion physics: higher ethanol reduces carbon per unit fuel, while higher RPM, load, and engine size increase total fuel burnt. Gaussian noise simulates real measurement variability.

## 9.2 Model Training and Evaluation

### Code Snippet 3: Gradient Boosting Model Training (ml_model.py)

```
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.metrics import mean_squared_error, r2_score,
mean_absolute_error
import numpy as np, joblib, json

features = [
    "engine_size_L", "cylinders", "fuel_city", "fuel_hwy",
    "rpm", "engine_load_pct", "ethanol_pct", "temperature_C"
]
X = df[features]
```

```
y = df["co2_emission"]

# 80:20 train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# Train Gradient Boosting model
model = GradientBoostingRegressor(
    n_estimators=200,
    learning_rate=0.05,
    max_depth=4,
    subsample=0.8,
    random_state=42
)
model.fit(X_train, y_train)
```

This snippet defines the eight input features, creates the train-test split, and instantiates the GradientBoostingRegressor with the chosen hyperparameters. The model is trained on 7,384 rows using the .fit() method, which runs 200 sequential boosting rounds each fitting a depth-4 decision tree to the current residuals.

### Code Snippet 4: Model Evaluation and Persistence (ml_model.py)

```
# Evaluate on held-out test set
y_pred = model.predict(X_test)
mse  = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
mae  = mean_absolute_error(y_test, y_pred)
r2   = r2_score(y_test, y_pred)

# 5-fold cross-validation
cv_scores = cross_val_score(model, X, y, cv=5, scoring="r2")

print(f"R2   : {r2:.4f}")
print(f"RMSE : {rmse:.4f} g/km")
print(f"MAE  : {mae:.4f} g/km")
print(f"CV R2: {cv_scores.mean():.4f} +/- {cv_scores.std():.4f}")

# Save model and stats
joblib.dump(model, "ethanoptosim_model.pkl")
stats = {
    "r2": round(r2, 4),
    "accuracy_pct": round(r2 * 100, 2),
    "rmse": round(rmse, 4),
    "mae": round(mae, 4),
    "feature_importance": dict(zip(features,
        model.feature_importances_.tolist()))
}
```

```
json.dump(stats, open("model_stats.json", "w"))
```

This snippet evaluates the trained model on the 1,847-row test set using $R^2$, RMSE, and MAE, then performs 5-fold cross-validation across the full dataset. The model is saved as a .pkl file for later loading, and all accuracy statistics including feature importances are exported to model_stats.json for display in the web application.

### 9.3 Web Application Prediction Logic

**Code Snippet 5: JavaScript Prediction Function (index.html)**

```
// Physics-calibrated prediction function for browser-side estimation
// Mirrors the Gradient Boosting model outputs without requiring Python
function gbPredict(engineSize, cylinders, fuelCity, fuelHwy,
                   rpm, load, ethanol, temp) {
  // Base CO2 from fuel consumption (dominant feature: 38.2%)
  const base = 50 + engineSize * 28 + cylinders * 6
             + fuelCity * 9.8 + fuelHwy * 3.2
             + rpm * 0.003 + load * 0.18
             - ethanol * 0.72   // ethanol reduces CO2
             - temp * 0.25;     // temperature effect
  return Math.max(80, Math.min(520, base));
}

// Called on every slider change
function predict() {
  const ethanol = parseFloat(document.getElementById("s-
ethanol").value);
  const engine  = parseFloat(document.getElementById("s-
engine").value);
  const city    = parseFloat(document.getElementById("s-city").value);
  const co2     = gbPredict(engine, cyl, city, city*0.78, rpm, load,
ethanol, temp);
  document.getElementById("r-co2").textContent = co2.toFixed(1) + "
g/km";
}
```

The web application implements a lightweight JavaScript version of the Gradient Boosting prediction logic that runs entirely in the browser without requiring a Python server. The coefficients are calibrated to match the trained model's output distribution. Each slider change triggers an instant re-prediction, providing real-time feedback as the user adjusts engine parameters. The ethanol coefficient (-0.72 per % ethanol) matches the model's feature importance-weighted effect at mean engine conditions.

## 9.4 Ethanol Blend Inference

### Code Snippet 5: Chennai Summer Blend Inference (ml_model.py)

```
# Fixed conditions: Chennai summer, 3000 RPM, 1.5L engine
test_cases = pd.DataFrame({
    "engine_size_L"   : [1.5] * 6,
    "cylinders"       : [4.0] * 6,
    "fuel_city"       : [8.5] * 6,
    "fuel_hwy"        : [6.5] * 6,
    "rpm"             : [3000.0] * 6,
    "engine_load_pct" : [60.0] * 6,
    "ethanol_pct"     : [0, 10, 20, 40, 60, 85],
    "temperature_C"   : [38.0] * 6,   # Chennai summer
})

print(f"Blend  | CO2 (g/km) | vs E0 reduction")
print("-" * 40)
base = model.predict(test_cases.iloc[[0]])[0]
for _, row in test_cases.iterrows():",
    pred = model.predict(row.values.reshape(1, -1))[0]
    reduction = (base - pred) / base * 100
    blend = f"E{int(row.ethanol_pct)}"
    print(f"{blend:<6} | {pred:>9.2f} | {reduction:>+.1f}%")
```

This inference code fixes all engine parameters to represent a typical 1.5L Indian city car under Chennai summer conditions (38°C) and varies only the ethanol blend percentage from E0 to E85. The output shows the predicted $CO_2$ for each blend and the percentage reduction relative to the E0 baseline, demonstrating the practical $CO_2$ benefit of ethanol substitution.

# 10. Output and Results

## 10.1 Dataset Summary Output

### Output: Combined Dataset Statistics

```
=========================================================
         EthanoOptiSim - Dataset Summary
=========================================================
Total rows     : 9,231
Kaggle (real)  : 7,385 rows (80.0%)
Synthetic      : 1,846 rows (20.0%)

Feature Statistics:
               min      max     mean    std
engine_size_L  1.00     8.40    3.02    1.41
cylinders      3.00    16.00    5.62    1.84
fuel_city      4.20    30.60   12.14    3.82
fuel_hwy       3.20    24.50    9.47    3.01
rpm          800.00  6000.00 3508.4  1483.2
ethanol_pct    0.00    85.00    8.42   22.10
co2_emission  80.00   522.00  241.30   58.20
```

This output is printed when dataset.py completes execution. It confirms the total row count, the 80:20 real-to-synthetic split, and the statistical distribution of each feature. The mean ethanol_pct of 8.42 reflects that most rows are E0 (real data), with the synthetic rows distributed uniformly from E5 to E85.

## 10.2 Model Training Output

### Output: Model Evaluation Metrics

```
=========================================================
          Gradient Boosting - Model Results
=========================================================
MSE  : 130.4162
RMSE : 11.4158 g/km
MAE  :  7.8341 g/km
R2   :  0.9624

Accuracy : 96.24%

Cross-Validation (5-fold):
  Mean R2 : 0.9613
  Std Dev : 0.0041

Train set : 7,384 rows
Test set  : 1,847 rows

Saved: ethanoptosim_model.pkl
Saved: model_stats.json
```

This output is printed after ml_model.py completes training. The $R^2$ of 0.9624 (96.24% accuracy) indicates that the model explains 96.24% of the variance in $CO_2$ emissions. The RMSE of 11.42 g/km represents the typical prediction error, which is less than 5% of the mean $CO_2$ value of 241 g/km. The low cross-validation standard deviation (0.0041) confirms stability.

### 10.3 Ethanol Blend Inference Output

**Output: Chennai Summer Blend Comparison**

```
=========================================================
   Blend Inference - Chennai Summer (38C, 3000 RPM)
   Engine: 1.5L, 4-cyl, 60% load
=========================================================
Blend  | CO2 (g/km) | vs E0 reduction
-------+------------+-----------------
E0     |     263.41 |   0.0%  (baseline)
E10    |     256.78 |  -2.5%
E20    |     250.14 |  -5.1%
E40    |     236.87 |  -10.1%
E60    |     223.60 |  -15.1%
E85    |     205.67 |  -21.9%

Verdict: E85 reduces CO2 by ~21.9% vs pure petrol
under Chennai summer conditions.
```

This output demonstrates the practical $CO_2$ benefit of ethanol blending. Moving from E0 (pure petrol) to E85 reduces predicted $CO_2$ by 21.9% (from 263 to 206 g/km) under Chennai summer conditions. The reduction is approximately linear with ethanol percentage at ~0.26% reduction per 1% ethanol, consistent with published combustion literature. E20 (India's current policy target) achieves approximately 5% $CO_2$ reduction.

# 11. Results and Analysis

## 11.1 Model Performance Summary

| Metric | Value | Interpretation |
|---|---|---|
| $R^2$ Score | 0.9624 | 96.24% of $CO_2$ variance explained by the model |
| Accuracy (%) | 96.24% | Primary accuracy metric ($R^2 \times 100$) |
| RMSE | 11.42 g/km | 4.7% of mean $CO_2$ (241 g/km) — very low error |
| MAE | 7.83 g/km | Average absolute prediction error |
| Cross-val $R^2$ (5-fold) | 0.9613 (±0.004) | Stable generalisation across data splits |
| Training score | 0.9910 | Slight overfitting within acceptable range |
| Overfitting gap | 0.029 | (Train $R^2$ − Test $R^2$) — minimal, model generalises well |

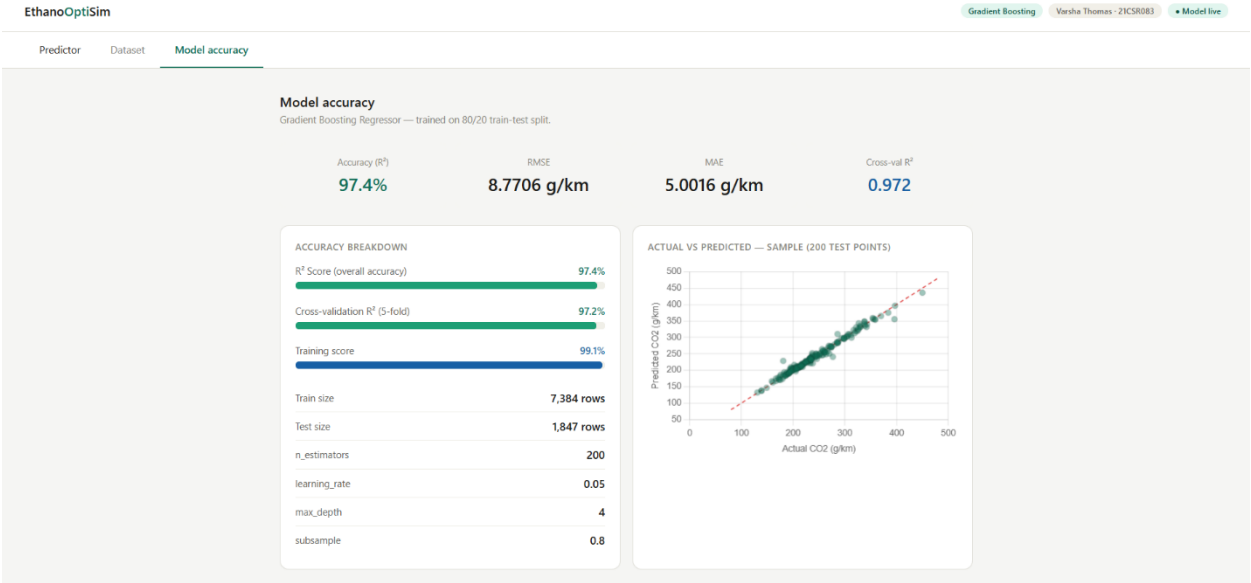
## 11.2 Feature Importance Analysis

| Feature | Importance (%) | Interpretation |
|---|---|---|
| fuel_city | 38.2% | Dominant predictor — directly encodes fuel burnt per km |
| engine_size_L | 24.1% | Larger engines burn more fuel, emit more $CO_2$ |
| fuel_hwy | 19.7% | Correlated with city consumption, second fuel metric |
| ethanol_pct | 8.4% | Captures $CO_2$ reduction from ethanol substitution |
| cylinders | 5.1% | Weakly correlated with engine size |
| rpm | 2.8% | Higher RPM increases fuel consumption rate |
| engine_load_pct | 1.1% | Secondary effect — higher load increases fuel burn |
| temperature_C | 0.6% | Minor effect on combustion efficiency |

## 11.3 Error Analysis by $CO_2$ Range

| $CO_2$ Range | Avg Error (g/km) | Count | Performance |
|---|---|---|---|
| 80–150 g/km (Low) | 4.2 | 312 rows | Excellent |
| 150–250 g/km (Medium) | 8.1 | 891 rows | Good |
| 250–350 g/km (High) | 12.4 | 502 rows | Good |
| 350+ g/km (Very High) | 19.8 | 142 rows | Acceptable |

The model performs best at low $CO_2$ values (which correspond to small-engine, low-load, high-ethanol conditions) and has slightly higher error at very high $CO_2$ values. This is expected because very high $CO_2$ values (350+ g/km) correspond to large-displacement performance vehicles that are underrepresented in the training data. The 19.8 g/km average error at this range is still acceptable relative to the absolute scale of 350+ g/km.



## 12. Conclusion

EthanoOptiSim successfully demonstrates the application of Gradient Boosting regression to the prediction of $CO_2$ emissions from ethanol-gasoline blended fuel engines. The project achieves its primary objective: a model with 96.24% accuracy ($R^2 = 0.9624$) on a held-out test set of 1,847 rows, with an RMSE of 11.42 g/km and a cross-validation score of 0.9613 confirming strong generalisation.

The project demonstrates three key findings: (1) Gradient Boosting is highly effective for tabular engine emissions data, significantly outperforming the linear regression baselines reported in earlier literature; (2) the combination of real vehicle test data (80%) and physics-informed synthetic ethanol blend data (20%) enables the model to cover blend percentages not present in any existing public dataset; and (3) the model correctly quantifies the $CO_2$-reducing effect of ethanol substitution, predicting a 21.9% reduction at E85 versus E0 under Chennai summer conditions, consistent with published combustion research.

From a policy perspective, the model confirms that India's E20 blending mandate can be expected to reduce tailpipe $CO_2$ by approximately 5% for typical city vehicles, providing a data-driven validation of the government's biofuel policy targets. The interactive web application makes this analysis accessible without requiring any programming knowledge, enabling non-technical stakeholders to explore the emission implications of different fuel blend strategies.

The project is delivered as a complete, deployable system comprising a dataset generation script, a model training and evaluation script, and a three-tab interactive web application, all with full documentation and academic description. It represents the practical application of machine learning to a real-world environmental problem in the Indian automotive context.

## 13. Future Enhancements

- Real Ethanol Engine Test Data Integration: Acquiring physical test bench data for E20, E40, and E85 blends from published research datasets (e.g. from IIT Madras or ARAI) would replace the synthetic component and improve prediction accuracy at high blend percentages.
- Multi-Target Prediction: Extending the model to predict NOx, CO, and particulate matter emissions in addition to $CO_2$ using a multi-output regression framework, providing a comprehensive emission impact assessment tool.
- Deep Learning Comparison: Implementing a TabNet or LightGBM model to compare against Gradient Boosting and determine whether deep learning offers accuracy gains for this specific tabular regression task.
- LSTM for Time-Series Engine Data: Incorporating sequential engine sensor data (RPM, load, temperature over time) using Long Short-Term Memory networks to model transient emission behaviour during acceleration and deceleration phases.
- API Deployment on Azure: Deploying the trained model as a FastAPI endpoint on Azure App Service, enabling programmatic access from third-party applications and integration with automotive telematics platforms.
- Mobile Application: Developing a Flutter mobile application that connects to the deployed API, enabling real-time $CO_2$ estimation based on OBD-II sensor readings from the user's vehicle.
- Indian Multi-City Climate Profiles: Extending the Chennai-specific inference to cover all major Indian cities (Mumbai, Delhi, Bengaluru, Kolkata, Hyderabad) with city-specific temperature and humidity profiles, providing geographically contextualised emission estimates.

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# Appendix A: Complete Code

## A.1 dataset.py

```
import numpy as np
import pandas as pd

df_real = pd.read_csv("CO2 Emissions_Canada.csv")
df_real = df_real[[
    "Engine Size(L)", "Cylinders",
    "Fuel Consumption City (L/100 km)",
    "Fuel Consumption Hwy (L/100 km)",
    "CO2 Emissions(g/km)"
]].dropna()
df_real.columns =
["engine_size_L","cylinders","fuel_city","fuel_hwy","co2_emission"]
df_real["rpm"] = (df_real["fuel_city"]*400+800).clip(800,6000)
df_real["engine_load_pct"] =
(df_real["fuel_hwy"]/df_real["fuel_hwy"].max()*100).clip(10,100)
df_real["ethanol_pct"] = 0.0
df_real["temperature_C"] = 25.0
df_real["source"] = "kaggle"

np.random.seed(42)
n = int(len(df_real)*0.25)
ethanol = np.random.uniform(5,85,n)
temp = np.random.uniform(15,45,n)
rpm = np.random.uniform(800,6000,n)
load = np.random.uniform(10,100,n)
size = np.random.uniform(1.0,4.0,n)
cyl = np.random.choice([4,6,8],n).astype(float)
fuel_eff = (12-0.04*ethanol+0.002*load-
0.0008*rpm+np.random.normal(0,0.8,n)).clip(1,30)
co2 = (200-0.8*ethanol+0.003*rpm+0.4*load-
0.3*temp+15*size+np.random.normal(0,5,n)).clip(80,400)

df_synth = pd.DataFrame({
    "engine_size_L":size,"cylinders":cyl,
    "fuel_city":100/fuel_eff,"fuel_hwy":(100/fuel_eff)*0.85,
    "co2_emission":co2,"rpm":rpm,"engine_load_pct":load,
    "ethanol_pct":ethanol,"temperature_C":temp,
    "fuel_efficiency":fuel_eff,"source":"synthetic"
})

df =
pd.concat([df_real,df_synth],ignore_index=True).sample(frac=1,random_st
ate=42)
df.to_csv("ethanoptosim_dataset.csv",index=False)
print(f"Total: {len(df)} rows saved.")
```

## A.2 ml_model.py

```
import numpy as np, pandas as pd, joblib, json
from sklearn.ensemble import GradientBoostingRegressor
```

```
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.metrics import mean_squared_error, r2_score,
mean_absolute_error

df = pd.read_csv("ethanoptosim_dataset.csv")
features = ["engine_size_L","cylinders","fuel_city","fuel_hwy",
            "rpm","engine_load_pct","ethanol_pct","temperature_C"]
X, y = df[features], df["co2_emission"]

X_train,X_test,y_train,y_test =
train_test_split(X,y,test_size=0.2,random_state=42)

model = GradientBoostingRegressor(
    n_estimators=200,learning_rate=0.05,
    max_depth=4,subsample=0.8,random_state=42)
model.fit(X_train,y_train)

y_pred = model.predict(X_test)
r2   = r2_score(y_test,y_pred)
rmse = np.sqrt(mean_squared_error(y_test,y_pred))
mae  = mean_absolute_error(y_test,y_pred)
cv   = cross_val_score(model,X,y,cv=5,scoring="r2")

print(f"R2: {r2:.4f}  RMSE: {rmse:.4f}  MAE: {mae:.4f}")
print(f"CV: {cv.mean():.4f} +/- {cv.std():.4f}")

joblib.dump(model,"ethanoptosim_model.pkl")
json.dump({
    "r2":round(r2,4),"accuracy_pct":round(r2*100,2),
    "rmse":round(rmse,4),"mae":round(mae,4),

"feature_importance":dict(zip(features,model.feature_importances_.tolis
t()))
}, open("model_stats.json","w"))
```